



PAPER – 2

ADVANCED FINANCIAL MANAGEMENT



QUESTIONS

Portfolio Management

1. Two friend Mr. A and Mr. N were discussing about the risks of market. While Mr. A is sort of risk averse, Mr. N is an aggressive investor and believes in taking risk.

Mr. N said we cannot diversify the market risk at all, and he quoted the Modern Portfolio Approach. Both friends analyze the market data for the few months and came out with expected returns on two stocks for a particular market.

Market Return	Aggressive	Defensive
7%	4%	9%
25%	40%	18%

Based on above scenario, answer the following questions:

- I. The Beta of Defensive stock is.....
 - (a) 2
 - (b) 0.5
 - (c) 4
 - (d) 1
- II. If the market return is equally likely to be 7% or 25% then expected return of Aggressive stock shall be.....
 - (a) 18%

- (b) 13.50%
 - (c) 22%
 - (d) 11%
- III. The Alpha of the Defensive stocks is.....
- (a) -10%
 - (b) 22%
 - (c) 5.50%
 - (d) 12%
- IV. The Modern Portfolio Theory was propounded by
- (a) William Sharpe
 - (b) Black Scholes
 - (c) Stephen Ross
 - (d) Harry Markowitz
- V. As per Capital Market Line (CML) Theory the Portfolios lying on the CML over the market portfolio are called
- (a) Lending Portfolios
 - (b) Borrowing Portfolios
 - (c) Diversified Portfolios
 - (d) Risk- Free Portfolios

Mutual Funds

2. Mr. X on 1.7.2021, during the initial offer of some Mutual Fund invested in 10,000 units having face value of ₹ 10 for each unit. On 31.3.2022, the dividend paid by the M.F. was 10% and Mr. X found that his annualized yield was 153.33%. On 31.12.2023, 20% dividend was given. On 31.3.2024, Mr. X redeemed all his balance of 11,296.11 units when his annualized yield was 73.52%.

Based on the above information answer the following questions:

- I. NAV per unit of the Fund as on 31.03.2022 shall be approximately.....
- (a) ₹ 19.50
 - (b) ₹ 20.50
 - (c) ₹ 21.50
 - (d) ₹ 22.50
- II. Total number of units as on 31.03.2022 shall be approximately.....
- (a) 10487.80 units
 - (b) 12585.65 units
 - (c) 9465.35 units
 - (d) 11575.40 units
- III. Total Dividend received by Mr. X as on 31.03.2023 shall be
- (a) ₹ 20,625.50
 - (b) ₹ 20,870.45
 - (c) ₹ 20,975.60
 - (d) ₹ 21,565.75
- IV. NAV per unit as on 31.03.2023 shall be approximately.....
- (a) ₹ 24.65
 - (b) ₹ 24.85
 - (c) ₹ 25.95
 - (d) ₹ 26.45
- V. NAV as on 31.03.2024 shall be approximately.....
- (a) ₹ 20.50

- (b) ₹ 25.95
- (c) ₹ 26.75
- (d) ₹ 27.20

Derivatives Analysis and Valuation

3. Mr. H is holding 100 equity shares of V Ltd. which is being quoted at ₹ 2,100 per share. He is interested in hedging downside risk of his holding as he is going to sell them after 2 months. A 2-month Call option is available at a premium of ₹ 60 per share and a 2-month put option is available at a premium of ₹ 50 per share. The strike price in both cases is ₹ 2,200. You are required to:
- (i) Suggest the position Mr. H should take in the option market to hedge his holding in the V Ltd.
 - (ii) Calculate his final position if after 2 months i.e., on the day of exercise the actual market price of per share of V Ltd. happens to be ₹ 2000, ₹ 2100, ₹ 2200, ₹ 2300 and ₹ 2400.
4. Mr. S has a portfolio of ₹ 50 lacs which he wants to invest in share market with rebalancing target after every 15 days to start with for a period of one month from now. The present NIFTY is 17025. The minimum NIFTY within a month can at most be 15322.50. He wants to know as to how he should rebalance his portfolio under the following situations, according to the theory of Constant Proportion Portfolio Insurance Policy, using "2" as the multiplier:
- (a) Immediately to start with.
 - (b) 15 days later-being the 1st day of rebalancing if NIFTY falls to 16321.89.
 - (c) 15 days further from the above date if the NIFTY touches 17512.14.

Note: 1. Assume that the value of his equity component will change in tandem with that of the NIFTY.

2. Round off calculations upto whole numbers.

Business Valuation

5. Compute EVA of X Ltd. with the help of following information:

Profit and Loss Statement	₹ lac	Balance Sheet	₹ lac
Revenue	1500	PPE	1500
Direct Costs	-585	Current Assets	450
Selling, General & Admin. Exp. (SGA)	-300		1950
EBIT	615	Equity	1050
Interest	-15	Reserves	150
EBT	600	Non-Current Borrowings	150
Tax Expense @30%	-180	Current Liabilities & Provisions	600
EAT	420		1950

Assume that Bad Debts provision of ₹ 30 Lac is included in the SGA, and same amount is reduced from the trade receivables in current assets.

Also assume that the pre-tax Cost of Debt is 12% and Equity shareholder's expected return is 10%.

Note: Make calculation in ₹ lac and round off up to 2 decimal points.

Foreign Exchange Exposure and Risk Management

6. PKR Ltd. has made purchases worth USD 8,00,000 on 1st May 2020 for which it has to make a payment on 1st November 2020. The present exchange rate is INR/USD 75. The company can purchase forward dollars at INR/USD 74. The company will have to make an upfront premium @ 1 per cent of the forward amount purchased.

The company can hedge its position with the following expected rate of USD in foreign exchange market on 1st May 2020:

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The company can hedge its position with the following expected rate of USD in foreign exchange market on 1st May 2020:

Exchange Rate	Probability
(i) INR/USD 77	0.15
(ii) INR/USD 71	0.25
(iii) INR/USD 79	0.20
(iv) INR/USD 74	0.40

You are required to advise the company for a suitable cover for risk assuming that the cost of funds to PKR Ltd. is 10 per cent per annum.

7. You as a dealer in foreign exchange have the following position in GBP on 31st October, 2019:

	GBP
Balance in the Nostro A/c Credit	2,00,000
Opening Position Overbought	1,00,000
Purchased a bill on London	1,60,000
Sold forward TT	1,20,000
Forward purchase contract cancelled	60,000
Remitted by TT	1,50,000
Draft on London cancelled	60,000

Decide the steps would you take, if you are required to maintain a credit Balance of GBP 65,000 in the Nostro A/c and keep as oversold position on GBP 20,000?

Advanced Capital Budgeting Decisions

8. JB Consultancy Group has determined relative utilities of cash flows of two forthcoming projects of its client company as follows:

Cash Flow in ₹	-150000	-100000	-40000	150000	100000	50000	10000
Utilities	-100	-60	-3	40	30	20	10

The distribution of cash flows of project X and Project Y are as follows:

Project X

Cash Flow (₹)	-150000	- 100000	150000	100000	50000
Probability	0.10	0.20	0.40	0.20	0.10

Project Y

Cash Flow (₹)	- 100000	-40000	150000	50000	100000
Probability	0.10	0.15	0.40	0.25	0.10

Which project should be selected and why?

Interest Rate Risk Management

9. B Bank Ltd. has entered into a plain vanilla swap through on Overnight Index Swap (OIS) on a principal of ₹ 10 crore and agreed to receive MIBOR overnight floating rate for a fixed payment on the principal. The swap was entered into on Monday, 10th July 2017 and was to commence on and from 11th July 2017 and run for a period of 7 days.

Respective MIBOR rates for Tuesday to Monday were:

8.75%, 9.15%, 9.12%, 8.95%, 8.98% and 9.15%.

If B Bank Ltd. received ₹ 4,170 net on settlement, calculate fixed rate and interest under both legs.

Notes:

- (i) Sunday is a holiday
- (ii) Work in rounded rupees and avoid decimal working
- (iii) Consider 365 days in a year.

Security Valuation

10. The Bank PK enters into a Repo for 9 days with Bank JJ in 6% Government Bonds 2022 for an amount of ₹ 20 crore. The other relevant details are as follows:

First Leg Payment (Start Proceed)	₹ 20,00,67,500
Second Leg Payment (Repayment Proceed)	₹ 20,03,17,590
Initial Margin	1.25%
Days of accrued interest	240

Assume 360 days in a year.

Calculate:

- (a) Repo Rate
- (b) Dirty Price and
- (c) Clean Price

11. Mr. A is holding 10000 shares of face value of ₹ 100 each of M/s. XYZ Ltd. He wants to hold these shares for long term and have no intention to sell.

On 1st January 2020, M/s. ABC Ltd. has made short sales of M/s. XYZ Ltd.'s shares and approached Mr. A to lend his shares under Stock Lending Scheme with following terms:

- (1) Shares to be borrowed for 3 months from 01-01-2020 to 31-03-2020,
- (2) Lending Charges/Fees of 1% to be paid every month on the closing price of the stock quoted in Stock Exchange and
- (3) Bank Guarantee will be provided as collateral for the value as on 01-01-2020.

Other Information:

- (a) Cost of Bank Guarantee is 8% per annum,
- (b) On 29-02-2020 M/s XYZ Ltd., declared dividend of 25%,
- (c) Closing price of M/s. XYZ Ltd.'s share quoted in Stock Exchange on various dates are as follows:

Date	Share Price in Scenario -1 Bullish	Share Price in Scenario -2 Bullish
01-01-2020	1000	1000
31-01-2020	1020	980
29-02-2020	1040	960
31-03-2020	1050	940

You are required to find out:

- (i) Earning of Mr. A through Stock Lending Scheme in both the scenarios,
- (ii) Total Earnings of Mr. A during 01-01-2020 to 31-03-2020 in both the scenarios,
- (iii) What is the Profit or loss to M/s. ABC by shorting the shares using through Stock Lending Scheme in both the scenarios?

Mergers and Acquisitions

12. L Ltd., is planning to acquire T Ltd., with the following data available for both the companies:

	L Ltd.	T Ltd.
Expected EPS	₹ 12	₹ 5
Expected DPS	₹ 10	₹ 3
No. of Shares	30,00,000	18,00,000
Current Market Price of Share	₹ 180	₹ 50

As per an estimate T Ltd., is expected to have steady growth of earnings and dividends to the tune of 6% per annum. However, under the new management the growth rate is likely to be enhanced to 8% per annum without additional investment.

You are required to:

- (i) Calculate the net cost of acquisition by L Ltd., if ₹ 60 is paid for each share of T Ltd.
- (ii) If the agreed exchange ratio is one share of L Ltd., for every three shares of T Ltd., in lieu of the cash acquisition as per (i) above, what will be the net cost of acquisition?
- (iii) Calculate Gain from acquisition.

International Financial management

13. The Management of a multinational company TL Ltd. is engaged in construction of Infrastructure Project. A proposal to construct a Toll Road in Nepal is under consideration of the Management.

The following information is available:

- ❖ The initial investment will be in purchase of equipment costing USD 250 lakhs. The economic life of the equipment is 10 years. The depreciation on the equipment will be charged on straight line method.
- ❖ EBIDTA to be collected from the Toll Road is projected to be USD 33 lakhs per annum for a period of 20 years.
- ❖ To encourage investment Nepalese government is offering a 15-year term loan of USD 150 lakhs at an interest rate of 6 per cent per annum. The interest is to be paid annually. The loan will be repaid at the end of 15 year in one tranche.
- ❖ The required rate of return for the project under all equity financing is 12 per cent per annum.
- ❖ Post tax cost of debt is 5.6 per cent per annum.
- ❖ Corporate Tax Rate is 30 per cent.
- ❖ All cash Flows will be in USD.

You are required to advise the management of TL Ltd. on the viability of the proposal by using Adjusted Net Present Value method. Ignore inflation.

Given

PVIFA (12%, 10) = 5.650, PVIFA (12%, 20) = 7.469, PVIFA (8%, 15) = 8.559, PVIF (8%, 15) = 0.315.

Note: Make calculations in USD Lakhs and round off them upto 3 decimal points.

Theoretical Questions

14. Explain the types of tests that can be employed to empirically verify the weak form of Efficient Market Theory.
15. Explain the concept of 'Unicorn'. Also mention the name of the startup became the India's first Unicorn.